

MATGEO Presentation: 2.10.82

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Problem Statement

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in three-dimensional space, where $\mathbf{u}$ and $\mathbf{v}$ are unit vectors which are not perpendicular to each other and

$$\mathbf{u} \cdot \mathbf{w} = 1, \quad \mathbf{v} \cdot \mathbf{w} = 1, \quad \mathbf{u} \cdot \mathbf{v} = 4.$$

If the volume of the parallelepiped, whose adjacent sides are represented by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, is $\sqrt{2}$, then the value of $|3\mathbf{u} + 5\mathbf{v}|$ is _____.
(2021)

Calculation

We are asked to find the magnitude of the vector $3\mathbf{u} + 5\mathbf{v}$. We will compute the square of its magnitude using the formula $\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{x}$.

$$\|3\mathbf{u} + 5\mathbf{v}\|^2 = (3\mathbf{u} + 5\mathbf{v})^\top (3\mathbf{u} + 5\mathbf{v}) \quad (3.1)$$

$$= 9(\mathbf{u}^\top \mathbf{u}) + 30(\mathbf{u}^\top \mathbf{v}) + 25(\mathbf{v}^\top \mathbf{v}) \quad (3.2)$$

We are given that $\mathbf{u}$ and $\mathbf{v}$ are unit vectors, which means $\mathbf{u}^\top \mathbf{u} = 1$ and $\mathbf{v}^\top \mathbf{v} = 1$. We are also given the value $\mathbf{u}^\top \mathbf{v} = 4$.

Final Result

Substituting these values formally:

$$||3\mathbf{u} + 5\mathbf{v}||^2 = 9(1) + 30(4) + 25(1) \quad (3.3)$$

$$= 9 + 120 + 25$$

$$= 154 \quad (3.4)$$

Taking the square root gives the magnitude:

$$|3\mathbf{u} + 5\mathbf{v}| = \sqrt{154} \quad (3.5)$$

Note on the Problem Statement

There appears to be a contradiction in the problem statement.

The vectors $\mathbf{u}$ and $\mathbf{v}$ are given as **unit vectors**. The dot product of two unit vectors is defined as:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cos \theta = (1)(1) \cos \theta = \cos \theta$$

Since the value of $\cos \theta$ must be in the range $[-1, 1]$, the given value $\mathbf{u} \cdot \mathbf{v} = 4$ is impossible.

Plot

Visualization of Parallelepiped and Vectors

